

MATHEMATICS

TARGET : JEE- Advanced 2023

CAPS-3**FUNCTION****Subjective Type Questions :**

- Find fundamental period of function $f(x) = 4 \cos^4\left(\frac{x-\pi}{4\pi^2}\right) - 2 \cos\left(\frac{x-\pi}{2\pi^2}\right)$
- Consider the function $f(x) = \frac{3x+a}{x^2+3}$ which has the greatest value $\frac{3}{2}$.
Find the minimum value of $f(x)$.
- Let f be one-one function with domain $\{x, y, z\}$ and range $\{1, 2, 3\}$. It is given that exactly one of the following statements is true and the remaining two are false.
 $f(x) = 1$; $f(y) \neq 1$; $f(z) \neq 2$. Determine $f^{-1}(1)$
- For what integral value of n , is 3π period of the function $\cos(nx) \sin\left(\frac{5x}{n}\right)$?
- Find the domain of definition of the function $f(x) = \frac{1}{[x-1] + [7-x] - 6}$, where $[.]$ denotes the greatest integer function

SCQ (Single Correct Type) :

- Consider the function $f(x) : A \rightarrow A$ which satisfy the condition $f(f(x)) = x$, then number of such functions for $A = \{1, 2, 3, 4, 5\}$ are
(A) 54 (B) 52 (C) 41 (D) 26
- Let $f(x) = \begin{cases} 2x+a, & x \geq -1 \\ bx^2+3, & x < -1 \end{cases}$ and $g(x) = \begin{cases} x+4, & 0 \leq x \leq 8 \\ -3x-2, & -2 < x < 0 \end{cases}$ $g(f(x))$ is not defined if
(A) $a \in (10, \infty)$, $b \in (5, \infty)$ (B) $a \in (4, 10)$, $b \in (5, \infty)$
(C) $a \in (10, \infty)$, $b \in (0, 1)$ (D) $a \in (4, 10)$, $b \in (1, 5)$
- For $x \neq 0, 1$ define $f_1(x) = x$, $f_2(x) = \frac{1}{x}$, $f_3(x) = 1-x$, $f_4(x) = \frac{1}{1-x}$, $f_5(x) = \frac{x-1}{x}$.
Let h be a function such that $f_3 \circ h \circ f_2 = f_4$ then h is equal to
(A) f_5 (B) f_4 (C) f_3 (D) f_2
- Let $f(x) = \max\{\sin t : 0 \leq t \leq x\}$
 $g(x) = \min\{\sin t : 0 \leq t \leq x\}$
and $h(x) = [f(x) - g(x)]$
where $[.]$ denotes greatest integer function, then the range of $h(x)$ is
(A) $\{0, 1\}$ (B) $\{1, 2\}$ (C) $\{0, 1, 2\}$ (D) $\{-3, -2, -1, 0, 1, 2, 3\}$

MCQ (One or more than one correct) :

10. Let $f : X \rightarrow Y$ where $3^{(f(x))^2} + 2^{-x} = 4$ and $f(x)$ is one-one and onto function and $f(x) \geq 0$ if
 (A) $X = (0, \infty), Y = (1, \sqrt{\log_3 4})$ (B) $X = [-\log_2 3, \infty), Y = [0, \infty)$
 (C) $X = (0, \infty), Y = (0, \sqrt{\log_3 4}]$ (D) $X = [-\log_2 3, \infty), Y = [0, \sqrt{\log_3 4}]$
11. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x+1) = \frac{f(x)-5}{f(x)-3} \forall x \in \mathbb{R}$. Then which of the following statement(s) is/are true _____.
 (A) $f(x+2) = f(x)$ (B) $f(x+4) = f(x)$ (C) $f(x+6) = f(x)$ (D) $f(x+8) = f(x)$
12. Let $f(x) = a_1 \cos(\alpha_1 + x) + a_2 \cos(\alpha_2 + x) + \dots + a_n \cos(\alpha_n + x)$. If $f(x)$ vanishes for $x = 0$ and $x = x_1$ (where $x_1 \neq k\pi, k \in \mathbb{Z}$), then _____.
 (A) $a_1 \cos \alpha_1 + a_2 \cos \alpha_2 + \dots + a_n \cos \alpha_n = 0$
 (B) $a_1 \sin \alpha_1 + a_2 \cos \alpha_2 + \dots + a_n \sin \alpha_n = 0$
 (C) $f(x) = 0$ has only two solutions $0, x_1$
 (D) $f(x)$ is identically zero $\forall x$
13. Which of the following function(s) have no domain?
 (A) $f(x) = \log_{x-1}(2 - [x] - [x]^2)$ where $[x]$ denotes the greatest integer function.
 (B) $g(x) = \cos^{-1}(2 - \{x\})$ where $\{x\}$ denotes the fractional part function.
 (C) $h(x) = \ln \ln(\cos x)$
 (D) $f(x) = \frac{1}{\sec^{-1}(\operatorname{sgn}(e^{-x}))}$
14. The values of x in $[-2\pi, 2\pi]$, for which the graph of the function $y = \sqrt{\frac{1+\sin x}{1-\sin x}} - \sec x$ and $y = -\sqrt{\frac{1-\sin x}{1+\sin x}} + \sec x$, coincide are
 (A) $\left[-2\pi, -\frac{3\pi}{2}\right] \cup \left(\frac{3\pi}{2}, 2\pi\right]$ (B) $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$
 (C) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (D) $[-2\pi, 2\pi] - \left\{\pm\frac{\pi}{2}, \pm\frac{3\pi}{2}\right\}$
15. Let $f(x) = [x]^2 + [x+1] - 3$, where $[x]$ denotes greatest integer less than or equal to x , then which of the following statement(s) is/are **CORRECT**?
 (A) $f(x)$ is many one function.
 (B) $f(x)$ vanishes for atleast three values of x .
 (C) $f(x)$ is neither even nor odd function.
 (D) $f(x)$ is aperiodic.

16. Let $f : I \rightarrow \mathbb{R}$, defined as $f(x) = 5 \cos 4\pi x - 13 \sin 7\pi x + 2$. Then which of the following alternative(s) is/are **TRUE**?
- (A) Range of f is a singleton set. (B) f is an even function.
 (C) $f(f(x)) = f(x) \forall x \in I$ (D) Inverse function of f is non existent.
17. Which of the following pair(s) of function have same graphs?
- (A) $f(x) = \frac{\sec x}{\cos x} - \frac{\tan x}{\cot x}$, $g(x) = \frac{\cos x}{\sec x} + \frac{\sin x}{\operatorname{cosec} x}$
 (B) $f(x) = \operatorname{sgn}(x^2 - 4x + 5)$, $g(x) = \operatorname{sgn}\left(\cos^2 x + \sin^2\left(x + \frac{\pi}{3}\right)\right)$ where sgn denotes signum function.
 (C) $f(x) = e^{\ln(x^2 + 3x + 3)}$, $g(x) = x^2 + 3x + 3$
 (D) $f(x) = \frac{\sin x}{\sec x} + \frac{\cos x}{\operatorname{cosec} x}$, $g(x) = \frac{2 \cos^2 x}{\cot x}$

Numerical based Questions :

18. N is the set of all natural number and $\mathbb{R}$ is the set of all real numbers. A function $f : N \rightarrow \mathbb{R}$ is given by:

$$f(n) = \frac{4n + \sqrt{4n^2 - 1}}{\sqrt{2n - 1} + \sqrt{2n + 1}}, \text{ then the value of } \left[\frac{\sum_{r=1}^{40} f(r)}{40} \right] \text{ is equal to } \underline{\hspace{2cm}}.$$

(where $[\cdot]$ represents greatest integer function)

19. If $f(x) = \begin{cases} -x + 1, & x \leq 0 \\ -(x-1)^2, & x \geq 1 \end{cases}$, then the number of solutions of the equation $f(x) - f^{-1}(x) = 0$ is /are _____.
20. $f(x)$ and $g(x)$ are linear function such that for all x , $f(g(x))$ and $g(f(x))$ are Identity functions. If $f(0) = 4$ and $g(5) = 17$, compute $f(136)$.
21. Let $f(x) = |x^2 - 9| - |x - a|$. Find the number of integers in the range of a so that $f(x) = 0$ has 4 distinct real root.

Comprehension Type Question:

Paragraph for question nos. 22 to 23

Let $f(x) = x^2 - 2x - 1 \forall x \in \mathbb{R}$. Let $f : (-\infty, a] \rightarrow [b, \infty)$, where 'a' is the largest real number for which $f(x)$ is bijective.

22. The value of $(a + b)$ is equal to
- (A) -2 (B) -1 (C) 0 (D) 1

23. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, then range of values of k for which equation $f(|x|) = k$ has 4 distinct real roots is
- (A) $(-2, -1)$ (B) $(-2, 0)$ (C) $(-1, 0)$ (D) $(0, 1)$

Paragraph for question nos. 24 to 26

An even periodic function $f : \mathbb{R} \rightarrow \mathbb{R}$ with period 4 is such that

$$f(x) = \begin{cases} \max. (|x|, x^2) & ; 0 \leq x < 1 \\ x & ; 1 \leq x \leq 2 \end{cases}$$

24. The value of $\{f(x)\}$ at $x = 5.12$ (where $\{ \}$ represents fractional part), is
 (A) $\{f(7.88)\}$ (B) $\{f(3.26)\}$ (C) $\{f(2.12)\}$ (D) $\{f(5.88)\}$
25. The equation of circle with centre lies on the curve $f(x)$ at $x = 9$ and touches x -axis, is
 (A) $x^2 + y^2 - 14x - 2y + 49 = 0$ (B) $x^2 + y^2 - 18x - 4y + 84 = 0$
 (C) $x^2 + y^2 - 18x - 2y + 81 = 0$ (D) $x^2 + y^2 - 18x + 2y + 81 = 0$
26. If $g(x) = |3\sin x|$, then the number of solutions of $f(x) = g(x)$ for $x \in (-6, 6)$, are
 (A) 5 (B) 7 (C) 3 (D) 9

Matrix Match Type :

27. Match the following :

Column-I		Column-II	
A	Let $f : [-1, \infty) \rightarrow (0, \infty)$ defined by $f(x) = e^{x^2+ x }$ then $f(x)$, is	P	one-one
B	Let $f : (1, \infty) \rightarrow [3, \infty)$ defined by $f(x) = \sqrt{10 - 2x + x^2}$, then $f(x)$ is	Q	into
C	Let $f : \mathbb{R} \rightarrow \mathbb{I}$ defined by $f(x) = \tan^5 \pi[x^2 + 2x + 3]$ where $[]$ denotes greatest integer function, then $f(x)$ is	R	many one
D	Let $f : [3, 4] \rightarrow [4, 6]$ defined by $f(x) = x-1 + x-2 + x-3 + x-4 $ then $f(x)$	S	onto
		T	periodic

28. Match the following :

Column-I		Column-II	
A	Let $f : \mathbb{R}^+ \rightarrow \{-1, 0, 1\}$ defined by $f(x) = \operatorname{sgn}(x - x^4 + x^7 - x^8 - 1)$ where sgn denotes signum function, then $f(x)$ is	P	Into
B	Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and satisfies $f(x) + x f(-x) = x + 1$, then $f(x)$ is	Q	One-one
C	Let $f : [0, 4] \rightarrow [0, 9]$ defined by $f(x) = \sqrt{6x - x^2}$, then $f(x)$ is	R	Many-one
D	Let $f : [0, 3] \rightarrow [2, 8]$ defined by $f(x) = 2^{ x-1 + x-2 }$, then $f(x)$ is	S	Even
		T	Onto

ANSWER KEY (CAPS - 03)

1. $2\pi^3$ 2. $\frac{-1}{2}$ 3. y 4. $n = \pm 1, \pm 3, \pm 5, \pm 15$
5. $x \in \mathbb{R} \sim (0, 1] \cup \{2, 3, 4, 5, 6, 7\} \cup (7, 8)$ 6. (D) 7. (A) 8. (B)
9. (C) 10. (AD) 11. (BD) 12. (ABD) 13. (ABCD)
14. (AC) 15. (ABCD) 16. (ABCD) 17. (ABCD) 18. (9.00)
19. (4.00) 20. (12) 21. (17) 22. (B) 23. (A)
24. (A) 25. (C) 26. (B)
27. (A) Q, R; (B) P, Q; (C) Q, R, T; (D) P, S 28. (A) P, R; (B) P, R, S; (C) P, R; (D) R, T